

Scoring Engine Specification

Project Name

CarbonWise

Version: 1.0

Status: Approved

1. Purpose

The Scoring Engine is the intelligence core of CarbonWise.

Its responsibility is to transform user activities into:

- Carbon Footprint Understanding
- Sustainability Tracking
- Personalized Reduction Opportunities

This engine directly powers the challenge objectives:

Understand

↓

Track

↓

Reduce

2. Design Principles

The scoring system must be:

Transparent

Users should understand why they received a score.

Actionable

Every score should lead to a recommendation.

Personalized

Different users should receive different recommendations.

Explainable

Users should understand:

- Their hotspots
 - Their opportunities
 - Their progress
-

3. Scoring Architecture

User Activities

↓

Carbon Calculation Engine

↓

Emission Aggregation

↓

Carbon Score Engine

↓

Hotspot Detection

↓

Recommendation Engine

↓

4. Carbon Categories

The scoring engine evaluates four primary emission categories.

Transportation

Examples:

- Car Travel
 - Motorcycle Travel
 - Flights
 - Taxi Usage
 - Public Transport
-

Food

Examples:

- Meat Consumption
 - Dairy Consumption
 - Vegetarian Meals
 - Vegan Meals
-

Energy

Examples:

- Electricity Usage
 - Heating
 - Cooling
-

Shopping

Examples:

- Clothing
 - Electronics
 - Household Purchases
-

5. Carbon Calculation Formula

Every activity is converted into emissions.

Formula

$$\begin{aligned} &\text{Emission} \\ &= \\ &\text{Activity Quantity} \\ &\times \\ &\text{Emission Factor} \end{aligned}$$

Example

Car Travel

$$\begin{aligned} &100 \text{ km} \\ &\times \\ &0.192 \text{ kg CO}_2/\text{km} \\ &= \\ &19.2 \text{ kg CO}_2 \end{aligned}$$

Example

Flight

500 km

×

0.255 kg CO₂/km

=

127.5 kg CO₂

6. Category Aggregation

Each category is calculated independently.

Transportation

Car

+

Bus

+

Train

+

Flight

Food

Meat

+

Dairy

+

Other Food Sources

Energy

Electricity

+

Heating

+

Cooling

Shopping

Clothing

+

Electronics

+

Household Purchases

7. Carbon Score Model

The platform converts emissions into a user-friendly score.

Score Range

0 - 100

Where:

100 = Excellent

0 = Extremely High Impact

Formula

Carbon Score

=

100

-

Normalized Emissions

Example

Monthly Emission

=

180 kg CO₂e

↓

Normalized

↓

Score = 68

8. Sustainability Rating System

The score is translated into understandable ratings.

Eco Champion

Score: 85 - 100

Meaning:

Very Low Carbon Impact

Sustainable

Score: 70 - 84

Meaning:

Good Sustainability Practices

Improving

Score: 50 - 69

Meaning:

Moderate Impact

High Impact

Score: 25 - 49

Meaning:

Critical

Score: 0 - 24

Meaning:

Immediate Improvement Needed

9. Hotspot Detection Engine

Purpose:

Identify the largest emission source.

Formula

$$\begin{aligned} &\text{Category \%} \\ &= \\ &\frac{\text{Category Emission}}{\text{Total Emission}} \\ &\times 100 \end{aligned}$$

Example

Transportation = 55%

Food = 20%

Energy = 15%

Shopping = 10%

Result

Primary Hotspot

=

Transportation

10. Recommendation Priority Engine

Purpose:

Determine which actions create the largest impact.

Priority Formula

Priority Score

=

Impact Potential

×

Feasibility

×

User Preference Alignment

Impact Potential

Measures:

How much CO₂ can be reduced

Feasibility

Measures:

How realistic the change is

Preference Alignment

Measures:

How compatible the action is
with user behavior

Example

Public Transport

Impact = 9

Feasibility = 8

Alignment = 7

Priority

=

504

11. Personalized Insight Engine

This is one of the most important parts of the project.

The challenge requires:

Personalized Insights

User A

Main Emission:

Transportation

Recommendation:

Use Public Transport

User B

Main Emission:

Food

Recommendation:

Reduce Meat Consumption

User C

Main Emission:

Energy

Recommendation:

Reduce Electricity Usage

12. Recommendation Difficulty Levels

Each recommendation receives a difficulty score.

Easy

Examples:

Turn Off Unused Lights

Walk Short Distances

Medium

Examples:

Use Public Transport

Reduce Meat Consumption

Hard

Examples:

Purchase Electric Vehicle

Install Solar Panels

13. Goal Progress Engine

Purpose:

Track measurable improvements.

Formula

$$\begin{aligned} \text{Progress \%} \\ &= \\ &\frac{\text{Current Reduction}}{\text{Target Reduction}} \\ &\times \\ &100 \end{aligned}$$

Example

Target:
100 kg CO₂

Current:
40 kg CO₂

Progress:
40%

14. Sustainability Improvement Score

Measures progress over time.

Formula

```
Improvement  
  
=  
  
Previous Emission  
  
-  
  
Current Emission
```

Example

```
Previous Month  
  
250 kg CO2  
  
Current Month  
  
180 kg CO2  
  
Improvement  
  
70 kg CO2
```

15. Recommendation Adoption Score

Measures user engagement.

Formula

$$\frac{\text{Accepted Recommendations}}{\text{Total Recommendations}} \times 100$$

Example

Accepted = 8

Generated = 10

Adoption Rate = 80%

16. Carbon Twin Forecast Engine

Purpose:

Predict future outcomes.

Inputs

- Current Activities
- Goals
- Recommendation History
- User Preferences

Output

Current Emissions

↓

Projected Emissions

↓

Potential Savings

Example

User Scenario:

Reduce Car Usage

Output:

Current

250 kg CO₂

Projected

180 kg CO₂

Savings

70 kg CO₂

17. Anti-Gaming Rules

Purpose:

Prevent unrealistic inputs.

Validation Rule 1

Reject impossible values.

Example:

50,000 km cycling in one day

Validation Rule 2

Reject negative values.

Example:

-200 kWh Electricity

Validation Rule 3

Reject duplicate submissions.

Example:

Same activity
submitted repeatedly

18. Performance Requirements

Supports:

Efficiency (Medium Impact)

Carbon Calculation

Target:

<100 ms

Recommendation Generation

Target:

<500 ms

Dashboard Load

Target:

<2 seconds

19. Testing Requirements

Supports:

Testing (Low Impact)

Unit Tests

Required For:

- Carbon Calculation Logic
 - Score Generation
 - Hotspot Detection
 - Recommendation Ranking
-

Integration Tests

Required For:

- Database Integration
 - API Integration
-

End-to-End Tests

Required For:

Calculator

↓

Score

↓

Recommendation

↓

Goal Creation

using:

Playwright

20. Success Metrics

The scoring engine is successful when:

- ✓ Users understand their score
- ✓ Hotspots are correctly identified
- ✓ Recommendations are personalized
- ✓ Progress is measurable

✓ Carbon Twin forecasts are meaningful

✓ Scores remain consistent

✓ Recommendation adoption improves over time

✓ The engine directly supports:

- Understand
- Track
- Reduce

objectives from the challenge statement.